

MOHAMMAD ZORAIZ

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EDUCATION

DUKE UNIVERSITY

Bachelor of Science, Electrical & Computer Engineering (AI/ML), Physics, and Computer Science (Triple Major)

Durham, NC
Aug 2023 - May 2027

Relevant Coursework: Data Structures & Algorithms; Computer Architecture; AI & Agents, Differential Equations, Probability, Signals & Systems, Physics E&M

SKILLS

Languages: Python, Java, C++, C, SQL, R, MATLAB, MIPS, x86, Assembly, TypeScript, Dart, JavaScript, Verilog, VHDL, Bash/Shell, HTML, CSS, SQL, Go
Technologies: AWS (Lambda, DynamoDB, S3, Bedrock, SageMaker), Microsoft Azure, Firebase (Genkit, Firestore, Auth), Supabase, Docker, Kubernetes, Linux, Git, React, Flutter, Scikit-learn, Pandas, NumPy, Qiskit, IBM Quantum Platform, AutoCAD, Autodesk Inventor, Adobe Illustrator & Photoshop, Figma, Kaggle
Machine Learning: PyTorch, TensorFlow, GraphQL, OpenCV, Hugging Face, Google App Engine, CNN, RNN, OpenAI Gym, Google Vertex AI, MLFlow

EXPERIENCE

Machine Learning Engineering Intern, Amazon Web Services

May 2025 - Aug 2025

- **AWS Agentic AI.** Developed platform adopted by Kiro, Amazon Q, and AWS Agentic AI teams to benchmark and accelerate iterative **LLM** improvement.
- Architected a serverless **Benchmark Analysis Platform**, automating large-scale code synthesis evaluation and accelerating **ML optimization** by 40%.
- Orchestrated 120+ distributed service interactions on **AWS** with **Lambda**, **DynamoDB**, and **S3**, delivering a fault-tolerant, highly scalable architecture.
- Developed advanced **algorithms** for cross-benchmark **pattern detection**, **AST-based dependency mapping**, and ML-driven **error fingerprinting**.
- Integrated **Amazon Bedrock** to power adaptive multi-armed bandit optimization (**RL**) and built a full **React** insights dashboard and modular test suite.

Software Engineering Intern, Amazon Web Services

May 2024 - Aug 2024

- Built statistical analysis pipelines in **Python**, **SQL**, **R** for **Amazon Q** Developer; boosted data throughput by 15% for reporting with **QuickSight**.
- Engineered **LLM pipeline analyzer**, optimizing latency and accuracy in **Lambda**; reduced error rates 15% across **LLM** transformations workloads.
- Optimized distributed LLM training pipelines with **SageMaker**, **Step Functions**, and **Redshift**, improving **inference** efficiency and model scalability 20%.
- Enhanced proficiency in scalable data engineering (**Redshift**), statistical modeling (**Athena**), and cloud ML infrastructure (**SageMaker**) on **AWS**.

Software Engineering Intern, Rockwell Automation

Jun 2023 - Aug 2023

- Developed **C++** firmware for **L8Z Motion Controllers**, leveraging **RTOS** scheduling and **EtherCAT** control loops to improve motion accuracy 20%.
- Authored technical white papers on experimental **motion libraries**, leveraging **MATLAB** simulations and **Git**-based **CI** for scalable **firmware design**.
- Optimized embedded build pipelines with **VMware**, **Docker**, and **Jenkins CI/CD**, reducing compilation latency and build times 35%.

Quantum Computing Researcher, MIT Lincoln Laboratory

Feb 2022 - Dec 2022

- Implemented **HHL algorithm** in **Microsoft Q#** and **Qiskit**, applying linear system solvers to optimize quantum **fluid-flow** and **cognition simulations**.
- Engineered quantum circuits leveraging **eigenvalue estimation**, amplitude amplification, and **Hamiltonian simulation**, achieving 33% faster runtimes.

Software Engineering Intern, Rockwell Automation

Jun 2022 - Aug 2022

- Developed automated **C++ verification suites** for **Logix Controllers** using **GoogleTest** and **VectorCAST**, improving debug fidelity and efficiency 25%.
- Streamlined **Studio5000 PLC controller** integration by engineering scalable firmware test harnesses with **Jenkins CI** and **Python** scripting.
- Architected verification pipeline integrating **containerized builds**, **hardware-in-loop testing**, and **regression analysis**, reducing validation time 30%.

PROJECTS

CtrlSlash: The Open-Source Google for AI IDEs

Jun 2025 - Current

- Developed ctrlslash, an **MCP**-driven documentation engine automatically scraping APIs, SDKs, frameworks, and dependencies, powered by **\$25k grant**.
- Built semantic search with **RAG**, **embeddings**, and **knowledge graphs**, enabling contextual doc injection and reducing framework **hallucinations** 80%.
- Engineered containerized AI dev environment with **Puppeteer** navigation, debugging hooks, and **IDE** integration across **Cursor**, **Windsurf**, and **Kiro**.

CipherShield: Making Discrimination Mathematically Impossible (won TreeHacks)

Feb 2025 - Current

- Engineered decentralized aggregation system using **OpenFHE** encryption and **Solidity** smart-contracts for mathematically guaranteed privacy.
- Implemented split-key **homomorphic encryption** with **C++** backend, enabling secure, auditable **blockchain**-verified protected-data aggregation.
- Optimized ciphertext **serialization**, key exchange, and large-payload management to ensure efficient, scalable **homomorphic encryption** operations.

ML Classifier for Gene Editing Signature Differentiation

Sep 2024 - Jan 2025

- Collaborated with **Life Edit Therapeutics (Novo Nordisk)** to engineer **Python**-based ML pipelines distinguishing edited from unedited gene expression profiles.
- Applied **EDA**, **feature selection**, and **PCA** to optimize classifier accuracy, enhancing large-scale **gene editing** analysis workflows, from **NIH** & **Duke** data.

SonicSync: Wearable Subaqueous Psychoacoustic Testing Apparatus

Aug 2023 - Aug 2024

- Co-developed with **Garmin** engineers a submersible, low-latency wearable in **AutoCAD** with **MATLAB**-driven **signal processing** algorithms.
- Implemented advanced **psychoacoustic** data acquisition and **wireless transmission** pipeline, enabling precise underwater auditory testing and analysis.
- Designed and implemented custom circuitry with **FPGA**-based control systems and **embedded integration**, culminating in **IEEE-published** research.

AchillesMath

Jun 2021 - Jun 2025

- Built adaptive K-7 math **app** in **Flutter/Python**, applying **cognitive load theory** and **scaffolding models** for hyper-personalized learning paths.
- Engineered **Firebase** pipelines with **reinforcement learning algorithms** and real-time analytics, winning Congressional App Challenge OH-14.

Computation-based Examination of Web-Based Student Learning Efficiency

Aug 2022 - May 2023

- Researched marginalized student learning using web-scraped assessments, applying **Scrapy**, **MATLAB**, and **computational psychiatry analytics**.
- Developed AcademicInsights, an open-source ML extension for analyzing web behavior in educational psychology research, used by Department of Education.

LANGUAGES

Native Proficiency: English, Urdu, & Hindi

Professional Working Proficiency: Arabic & Spanish

Basic Proficiency: Turkish, ASL

ACHIEVEMENTS

Amazon Future Engineer Scholar; Amazon, 1/250 Scholars Internationally, awarded \$40k, for international excellence in engineering and CS

Mar 2023

Lockheed Martin STEM Scholar; Lockheed Martin, 1/100 Scholars Nationally, awarded \$40k, for national excellence in engineering and innovation.

Apr 2023

Dominion Energy Educational Equity Scholar; Dominion Energy, 1/40 Scholars Nationally, awarded \$40k, for exceptional national social impact.

Apr 2023

Congressional Commendation; Congressman David Joyce, for outstanding dedication to NE Ohio community advancement in childhood education.

Jun 2022

ACTIVITIES

- **Dark Matter & AI Research (Kotwal Labs):** Leveraging AI and Integrated Circuitry (**FPGA**) to find Dark Matter. High Energy Physics.
- **ProductSpace@Duke (Fellowship Director):** Lead product fellows in technical **product management** education and mentor multidisciplinary projects teams.
- **HackDuke (Organizer):** Plan and execute HackDuke, coordinating logistics, sponsors, teams, and fostering tech-driven social impact.